

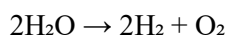
EXPERIMENT 4: Water Splitting Experiment

Electrolysis is a process by which a chemical reaction is carried out using an electric current. In water electrolysis, water is oxidized at the anode (negative) to produce oxygen and reduced at the cathode (positive) to produce hydrogen. Hydrogen is used to power fuel-cell cars and trucks. Electrolysis is one of the cheapest and cleanest ways to produce pure hydrogen gas. The efficiency of the water electrolysis highly depends on the electrodes and the electrolytes used in this process.

Materials: Copper Wire, Graphite Rods, graduated burette, Battery (9V PP3), Beaker, Electrolytes.

Experimental Procedure:

1. Connect the electrodes to the power source.
2. Place the electrodes in a deionized water-filled inverted burette in a Beaker.
3. Apply voltage to initiate electrolysis. Hydrogen gas evolves at the cathode (negative electrode), while oxygen gas evolves at the anode (positive electrode).
4. The chemical reaction is:



5. Bubbles should form at each electrode - hydrogen at the cathode and oxygen at the anode collected at the top of the inverted measuring cylinders.
6. Repeat the procedure with tap water, NaCl solution, and MgCl₂ solution.

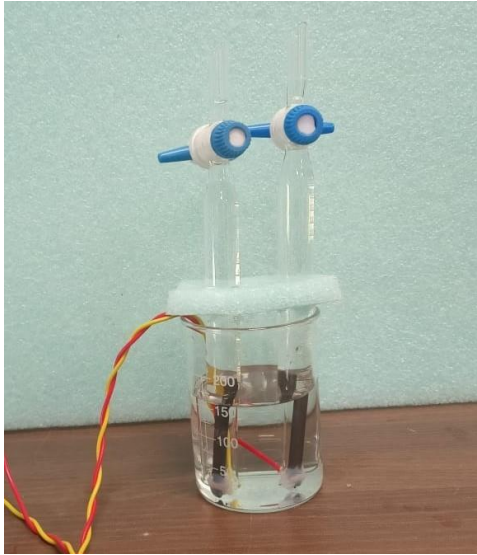
Preparation of NaCl and MgCl₂ solutions:

Prepare 1 M solution of NaCl and MgCl₂.

Dissolve 15 - 20 mL of 1 M solution into 150 mL of deionized water to make an electrolyte solution.

Observations:

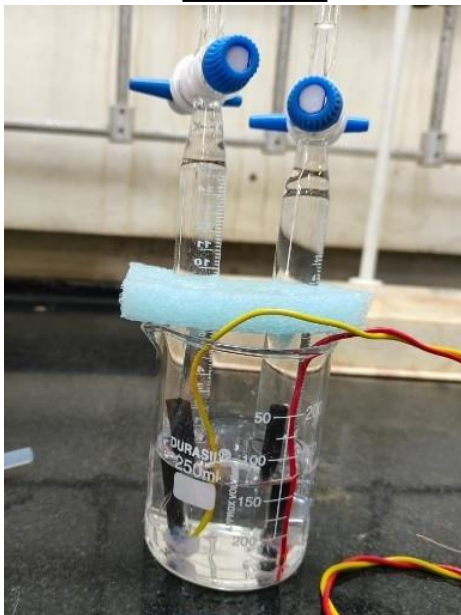
1. A very small amount of hydrogen bubbles will be observed in deionized water but the amount of hydrogen will increase after the addition of electrolytes.



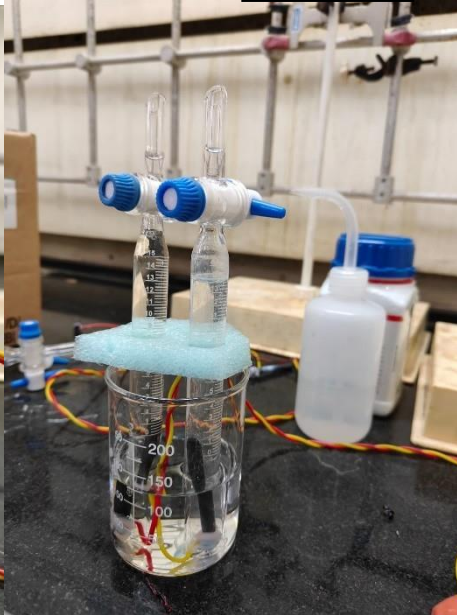
DI water



Tap water



NaCl electrolyte



MgCl₂ electrolyte